

ECE 434: DSP

Problem Set 1

Pranav Chandra N. V

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1. Determine the mod-4 and mod-4 reductions of the length 8 signal vector:

$$x = [1, 2, -2, 3, 4, -2, -1, 1]^T \quad (1)$$

Solution:

Mod-N reduction involves partitioning the vector into sequences of length N and then summing the sequences element-wise.

Thus, mod-4 reduction of x is given by:

$$x_{mod4} = [1 + 4, 2 - 2, -2 - 1, 3 + 1]^T = [5, 0, -3, 4]^T \quad (2)$$

Similarly, mod-3 reduction of x is given by:

$$x_{mod2} = [1 + 3 - 1, 2 + 4 + 1, -2 - 2]^T = [3, 7, -4]^T \quad (3)$$

Note that for the mod-3, the third sequence has only 2 elements, so we assume the third element is a 0 (pad zeroes where required).

2. Let $x = [1, 2, 2, 1, 2, 1, 1, 2]$, compute the 4-point DFT using the matrix form definition

Solution:

$$\begin{bmatrix} 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ 1 & -j & -1 & j & 1 & -j & -1 & j \\ 1 & -1 & 1 & -1 & 1 & -1 & 1 & -1 \\ 1 & j & -1 & -j & 1 & j & -1 & -j \end{bmatrix} \times \begin{bmatrix} 1 \\ 2 \\ 2 \\ 1 \\ 2 \\ 1 \\ 1 \\ 2 \end{bmatrix} = \begin{bmatrix} 12 \\ 0 \\ 0 \\ 0 \end{bmatrix} \quad (4)$$

Now, compute the mod-4 reduction of x and then compute the 4-point DFT of the reduced signal. Compare the results with the previous DFT.

Solution:

Mod-4 reduction of x is given by:

$$x = [3, 3, 3, 3] \quad (5)$$

Now, computing the 4-point DFT of the reduced signal:

$$\begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & -j & -1 & j \\ 1 & -1 & 1 & -1 \\ 1 & j & -1 & -j \end{bmatrix} \times \begin{bmatrix} 3 \\ 3 \\ 3 \\ 3 \end{bmatrix} = \begin{bmatrix} 12 \\ 0 \\ 0 \\ 0 \end{bmatrix} \quad (6)$$

This question basically shows that to compute reduced points DFT, taking the mod-N version of the main sequence provides you with a more efficient way of computing the DFT.

3. Compute the 8 point FFT of a length 8 signal which are 8 samples of the signal:

$$x[n] = 4 \cos\left(\frac{\pi n}{2}\right) + \cos(\pi n) \quad (7)$$

Discuss if the FFT accurately represents the entire spectrum of n .

Solution:

The 8 points sequence to be taken is given by:

$$x = [0, 0, 16, 0, 8, 0, 16, 0] \quad (8)$$

Now, computing the 8-point DFT of the signal: